

The eddy diffusion coefficient K_{zz} : theory, empirical determinations, and what fixes it at the LHS 1140 b wind base

1. Why this memo exists, and the answer up front

EXHALE's binary H/He element-diffusion operator (`src/modules/functions/binary_element_diffusion.f90`, formulated in `docs/binary_diffusion_design.md`) carries one number that the code cannot derive from anything it solves: the eddy diffusion coefficient `He_Kzz`. On LHS 1140 b that number moves the He I 10830 equivalent width by four orders of magnitude and the escape rate by 0.23 dex (`LHS1140b/kzz_decision.md`, section 3), so it is not a detail. This memo collects what the physics, the modelling literature and the measurements actually say about K_{zz} , and asks what of it survives contact with a 226 K, 1 μ bar base on a temperate super-Earth.

Summary. K_{zz} is a one-dimensional closure, not a material property. It is defined by the ratio of a horizontally averaged tracer flux to a horizontally averaged mixing-ratio gradient, so it is a property of the pair (atmosphere, tracer) and not of the atmosphere alone; Zhang & Showman (2018a,b) show that different species in the same atmosphere have different K_{zz} profiles and that the inferred value can even be negative when transport is not diffusive. Three theoretical constructions dominate: a convective mixing-length branch ($K = w_* L$, deep and irrelevant at 1 μ bar), a breaking-gravity-wave branch that supplies the familiar $K \propto P^{-1/2}$ growth (Lindzen 1981), and effective coefficients read off three-dimensional tracer transport in GCMs (Parmentier et al. 2013; Charnay et al. 2015). Empirical determinations come from solar-system homopause occultations (Festou et al. 1981: $1.4 \times 10^6 \text{ cm}^2 \text{ s}^{-1}$ at Jupiter's homopause), from chemical quenching (Prinn & Barshay 1977, the founding case), and, on exoplanets, from cloud and haze lofting arguments that give lower bounds only.

Two conclusions matter for the present application, and both are negative. First, *every* published number that can be evaluated at 1 μ bar comes from an atmosphere warmer and more strongly forced than this one, and the only scaling that has ever been evaluated with LHS 1140 b's own gravity and temperature (Arfaux & Lavvas 2023) lands one to two decades below the hot-Jupiter practice the code currently inherits. Second, and less obvious, a *constant* K_{zz} is the wrong shape of assumption for this problem regardless of its magnitude: the molecular coefficient it competes against rises by nearly five decades between the base and $1.2R_p$, so no single constant can both mix the base and follow the molecular coefficient outward, while Lindzen's own result is that the eddy coefficient should *turn over* rather than keep growing. The adopted `He_Kzz` = $10^9 \text{ cm}^2 \text{ s}^{-1}$ is defensible only as a stated convention inherited from hot-Jupiter work, and it should be reported alongside the 10^8 – 10^{10} bracket, not instead of it.

This memo does not repeat the survey. The verification status of every published number quoted here is in `LHS1140b/kzz_literature.md`; the D_{eff} structure, the `He_Kzz` scan and the adopted value are in `LHS1140b/kzz_decision.md`; the operator and its $(D+K)$ form are in `docs/binary_diffusion_design.md`, sections 2.3 and 9.4. What is new here is the physics behind the numbers, four newly verified primary citations (Sect. 6.), and an assessment of what a principled determination would require.

2. What K_{zz} is

2.1 The closure

Take a three-dimensional atmosphere, a tracer with mixing ratio q , and average horizontally over a level surface. The horizontally averaged vertical tracer flux contains a mean-flow part and an eddy correlation $\overline{\rho w'q'}$. The eddy diffusion closure asserts that the second is proportional to the mean gradient,

$$\overline{\rho w'q'} = -\bar{\rho}K_{zz}\frac{d\bar{q}}{dz}, \quad \text{i.e.} \quad K_{zz} \equiv -\frac{\overline{\rho w'q'}}{\bar{\rho}d\bar{q}/dz}, \quad (1)$$

which is a *definition* of K_{zz} whenever both sides are measurable, and a *model* only when K_{zz} is asserted in advance. Everything in Sect. 3. is an attempt to predict the right-hand side without running the three-dimensional flow.

Two features of Eq. (1) are worth stating explicitly because they are what separate eddy from molecular diffusion.

The gradient is in mixing ratio, not density. A uniformly mixed atmosphere has $d\bar{q}/dz = 0$ and therefore zero eddy flux, however steep its density gradient. Molecular diffusion, by contrast, drives each species toward its *own* scale height and so has a non-zero flux in a uniformly mixed atmosphere. Eddy mixing transports the mixture as a whole; molecular diffusion separates it. This is why the aeronomical transport equation (Banks & Kockarts 1973) carries $D+K_{zz}$ on the gradient term and D alone on the settling term, which is exactly the form EXHALE implements (binary_diffusion_design.md, eq. 4):

$$J_{\text{total}} = J_{\text{mol}} - \rho K_{zz} \frac{dX}{dr}, \quad (2)$$

with X the helium mass fraction and J_{mol} the binary molecular flux of the design memo's eq. (2)–(3).

The homopause is where the two are equal. Setting $K_{zz} = D$ defines the homopause. Below it the composition is held at the reservoir value; above it the elements separate on their own scale heights. The homopause is therefore not a property of K_{zz} alone but of the crossing of two profiles, and where a model places it is the single most consequential thing K_{zz} does. In the LHS 1140 b runs the homopause radius moves from below the base at $K_{zz} = 0$ to $1.44R_p$ at $K_{zz} = 10^{11}$ (kzz_decision.md, section 3).

2.2 K_{zz} is a property of the (atmosphere, tracer) pair

Equation (1) has a tracer in it. Zhang & Showman (2018a,b) develop this into the central theoretical caveat of the subject: the effective K_{zz} inferred from a tracer depends on that tracer's chemical lifetime, on the strength of the large-scale circulation and on horizontal eddy mixing, so that “different chemical species in a single atmosphere should in principle have different eddy diffusion profiles”, and, where the transport is not diffusive at all — a mean overturning circulation, for instance, whose upwelling and downwelling branches carry different tracer abundances — the K_{zz} that reproduces the mean profile can be *negative*.

The consequence for us is specific and is not addressed anywhere in the literature surveyed. The tracer EXHALE diffuses is the elemental He/H ratio: chemically inert, of unbounded lifetime, and subject to a settling velocity of its own. A K_{zz} calibrated on CO/CH₄ quench chemistry, or on a settling condensate, is calibrated on a tracer with a finite lifetime; in Zhang & Showman's framework a longer-lived tracer generally samples the circulation more completely

and yields a *different* effective coefficient. Whether the elemental ratio should take a larger or smaller K_{zz} than the quench tracers is not something this memo can settle; it is recorded as a systematic that the transfer of literature values into Eq. (2) silently assumes away.

2.3 Definitional slack

Three further ambiguities affect numbers taken from print:

- *Which mixing ratio.* Eq. (1) written for a molar mixing ratio and for a mass fraction differ when the mean molecular weight varies with height — exactly the situation above a homopause. EXHALE’s eddy term mixes toward uniform *mass* fraction (parameters.f90, the he_kzz comment).
- *Units.* $1 \text{ m}^2 \text{ s}^{-1} = 10^4 \text{ cm}^2 \text{ s}^{-1}$, and the literature uses both. kzz_literature.md, section 2.1 documents a place where the two appear to have been confused in print.
- *Where it is quoted.* A value is meaningless without the pressure at which it applies, unless the source asserts a plateau. Half the disagreement in Table 2 is a disagreement about height, not about magnitude.

3. Theory

3.1 Convective mixing length

In a convecting layer the natural estimate is $K_{zz} \simeq w_* L$ with w_* a convective velocity scale and L a mixing length. Writing w_* in terms of the convective heat flux gives the free-convection form of Gierasch & Conrath (1985), which Ackerman & Marley (2001) adopt as their eq. (5):

$$K = \frac{H}{3} \left(\frac{L}{H} \right)^{4/3} \left(\frac{RF}{\mu \rho_a c_p} \right)^{1/3}, \quad w_* = K/L, \quad (3)$$

with $H = RT/\mu g$ the scale height, F the convective heat flux, ρ_a the density and c_p the specific heat. They state that the coefficient K “is assumed to be the same as that for heat as derived for free convection”, that “the mixing length is typically assumed to be the pressure scale height” in freely convecting regions, and that w_* is “the convective velocity scale from mixing-length theory”. The same expression is the one Charnay et al. (2015) write as their eq. (16).

Three properties of Eq. (3) carry into everything downstream.

It is weakly sensitive to the energetics and strongly sensitive to the scale height. $K \propto F^{1/3}$ and $K \propto H$ at fixed L/H . A factor 10^3 in heat flux buys one decade in K ; a factor 10 in scale height buys one decade as well. For a cold, high-gravity, helium-dominated atmosphere both push the same way and the scale height pushes harder.

Its normalization is admitted to be free. Ackerman & Marley adopt the prefactor $1/3$ “based on previous modeling studies of the Jovian atmosphere” and state plainly that “the constant coefficient scaling the eddy diffusion coefficient is only loosely constrained by observations”. Their baseline Jovian values just below the ammonia cloud are $H = L = 20 \text{ km}$, $K = 2 \times 10^8 \text{ cm}^2 \text{ s}^{-1}$ and $w_* = 1.1 \text{ m s}^{-1}$; they also impose a floor $K_{\min} = 10^5 \text{ cm}^2 \text{ s}^{-1}$, writing that “we also

assume an eddy diffusion coefficient no less than a prescribed minimum value”, which the paper attributes to “residual turbulence” from “breaking buoyancy waves”, citing Lindzen (1981). (Those numerals and the attribution are read from the arXiv version; the sentences quoted are confirmed present in the published text, and the two fragments in the last clause are individually confirmed — see Sect. 6.)

It is a deep-atmosphere statement, and the flux in it is the internal one. Convection carries the interior heat flux; the absorbed stellar flux is re-radiated from the photosphere and does not pass through the convective zone. Ackerman & Marley evaluate $F = \sigma T_{\text{eff}}^4$ with Jupiter’s *internal* $T_{\text{eff}} = 124$ K. Charnay et al. (2015) warn that this formula “strongly overestimates the convective heat flux for irradiated planets” when F_c is set from T_{eff} . This corrects a loose step in LHS1140b/kzz_literature.md, section 4, which compares LHS 1140 b and HD 209458 b through their *insolation* on the convective branch: the insolation ratio bears on the wave-driving branch of Sect. 3.2, through the day/night contrast, and not on Eq. (3). The direction of the conclusion in that document is unaffected — both branches point the same way — but the convective comparison should be made through the internal heat flux, which is unconstrained for LHS 1140 b and presumably small for an old, low-mass, temperate planet.

And it does not apply at the wind base. The LHS 1140 b base is at $1 \mu\text{bar}$ in a stably stratified, radiatively controlled layer, some 80–90 isothermal scale heights above the 1 bar level ($H = kT / \mu m_H g = 25$ km at $T = 226$ K, $\mu = 4$, $g = 1837 \text{ cm s}^{-2}$; $H/R_p = 2.3 \times 10^{-3}$). Equation (3) sets the *deep* value on which the wave branch of the next section is built; it says nothing directly about the base.

3.2 Breaking gravity waves and the $P^{-1/2}$ growth

The mechanism that supplies mixing in a stably stratified upper atmosphere is wave breaking, and its canonical treatment is Lindzen (1981). The argument in outline: an internal gravity wave propagating upward conserves its vertical flux of wave energy, so its velocity amplitude grows as $\rho^{-1/2}$, i.e. roughly as $P^{-1/2}$; the growth cannot continue indefinitely, because at large enough amplitude the wave-induced shear or lapse rate makes the flow unstable and the wave breaks; the turbulence produced by breaking is then parameterized as an eddy coefficient of exactly the size needed to hold the wave at its saturation amplitude.

Lindzen’s abstract states the premise in those terms — the expectation that “sufficient turbulence would be generated to prevent the growth of wave amplitude with height (roughly as (basic pressure) $^{-1/2}$)” — and the paper’s stated purpose is to extend that picture to smaller-amplitude waves, to the effect of mean winds on the waves, and to the waves’ effect on the mean momentum budget.

Two things follow, and the second is usually dropped.

The slope. A saturated wave spectrum gives an eddy coefficient that *grows* with height, and the $\rho^{-1/2}$ amplitude growth is where the exponent comes from. Every power-law $K_{zz}(P)$ in the exoplanet literature is of this family: Parmentier et al. (2013) fit $K_{zz} = 5 \times 10^4 P_{\text{bar}}^{-1/2} \text{ m}^2 \text{ s}^{-1}$ to their HD 209458 b GCM tracers; Charnay et al. (2015) fit $K_{zz} = K_{zz,0} P_{\text{bar}}^{-0.4} \text{ m}^2 \text{ s}^{-1}$ on GJ 1214 b and justify the exponent by noting that planetary waves “generally grow as $P^{-1/2}$ ”, expecting “an eddy diffusion coefficient with an exponent between $-1/3$ and $-1/2$ ”.

The turnover. Lindzen’s own conclusion is that the eddy coefficient does *not* rise without limit. The mean wind filters the wave spectrum before it arrives, critical levels remove waves, and the paper reports that “eddy coefficients may peak” at particular altitudes and then “decrease with height rather sharply above these levels” (terrestrial mesosphere; the abstract quotes 50 km in winter mid-latitudes and 70–80 km in summer). Extrapolating a saturated-wave power law upward past the level where the wave field is exhausted is therefore not supported by the theory that produced the power law. This is the physical content of the plateau assumption used by Moses et al. (2011) — constant K_{zz} above the GCM top — and of the homopause plateau of Arfaux & Lavvas (2023) in Sect. 3.4, and it is the reason the extrapolation of Charnay’s $P^{-0.4}$ fit 1.5 decades above its GCM top (`kzz_literature.md`, section 3) should be read as an upper bound.

And the magnitude is set by the source, not by the slope. Wave breaking fixes how K_{zz} varies with pressure; what fixes its value is how much wave flux is launched from below. On tidally locked, strongly irradiated planets that source is the day/night thermal contrast — which is what Parmentier et al. (2013) identify, finding that “vertical mixing results not from small-scale convection but from the large-scale circulation driven by the day-night heating contrast”. LHS 1140 b’s day/night contrast, in absolute terms, is smaller than a hot Jupiter’s by orders of magnitude, so the same mechanism should be correspondingly weaker even if the slope survives. This is the branch on which the insolation comparison of `kzz_literature.md`, section 4 is the relevant one.

3.3 Effective diffusion measured in a GCM

The most defensible route to K_{zz} for a given planet is to stop predicting it and to measure it in a three-dimensional model, through Eq. (1). The methodology, as used by Parmentier et al. (2013) and Charnay et al. (2015), is:

1. Run a GCM to a statistically steady circulation.
2. Advect one or more *passive* tracers in that flow. Passive means the tracer does not feed back on the dynamics; it may still carry a source/sink or, in Charnay’s case, a settling velocity representing condensate particles of a given size.
3. Average the tracer field horizontally at each level to get $\bar{q}(z)$ and the eddy flux $\overline{\rho w' q'}(z)$.
4. Form the flux/gradient ratio of Eq. (1), or equivalently fit a one-dimensional diffusion model to reproduce $\bar{q}(z)$, and report the result as a power law in pressure.

Two results of that exercise are more important than the fitted constants.

First, the effective coefficient is much smaller than the naive estimate. Parmentier et al. report their GCM-derived K_{zz} to be “two orders of magnitude smaller than what is obtained when multiplying the vertical scale height by the root mean square of the vertical velocity” — the $w_{\text{rms}}H$ estimate. The reason is that $w_{\text{rms}}H$ counts all vertical motion as mixing, including the reversible up-and-down of a large-scale circulation that returns the tracer to where it came from. The flux/gradient ratio counts only the part that is correlated with the tracer anomaly.

Second, and following from it, the tracer matters (Sect. 2.2): Charnay’s coefficients depend on the settling velocity of the particle the tracer represents, and are quoted as separate fits for different atmospheric compositions.

Neither GCM covers a 1 μ bar level. Charnay’s model top is 3 Pa (3×10^{-5} bar); Parmentier’s parameterization is stated valid “between ~ 1 bar and ~ 1 microbar”, which is the only entry in the survey whose stated validity reaches the pressure of interest.

3.4 Homopause scaling

The construction closest in spirit to what a wind model needs is a prediction of the coefficient *at the homopause*, taken constant above it. Arfaux & Lavvas (2023) build a three-branch parameterization — a deep convective branch, a gravity-wave middle branch, and a constant homopause plateau — and write the plateau value as a scaling from Jupiter:

$$K_{\text{top}} = K_{zz}^J \frac{v}{v_J} \frac{g_J}{g} \frac{T}{T_J}, \quad K_{zz}^J = 10^7 \text{ cm}^2 \text{ s}^{-1}, \quad (4)$$

with v a characteristic turbulent velocity ($v/v_J = 5$ retained for HD 209458 b). The T/g dependence is transparent once read as $K \sim vH$: the mixing length at the homopause is the scale height, $H = kT/\mu m_H g$, so at fixed composition $K_{\text{top}} \propto vT/g$ directly. (That reading is mine, not a statement quoted from the paper.)

Evaluated with LHS 1140 b’s own $g = 18.4 \text{ m s}^{-2}$ and $T = 226 \text{ K}$ against Jupiter’s $g_J = 24.79 \text{ m s}^{-2}$, $T_J = 110 \text{ K}$ (kzz_literature.md, section 3), Eq. (4) gives $2.8 \times 10^7 \text{ cm}^2 \text{ s}^{-1}$ for $v/v_J = 1$ and 1.4×10^8 for $v/v_J = 5$. The same formula applied to HD 209458 b ($g = 9.4 \text{ m s}^{-2}$, $T = 1450 \text{ K}$, $v/v_J = 5$) returns 1.7×10^9 , within a factor 2 of the $10^9 \text{ cm}^2 \text{ s}^{-1}$ that Taylor et al. (2025) adopt on the strength of this parameterization, so the scaling reproduces the hot-Jupiter practice it is meant to reproduce and its much lower LHS 1140 b prediction is not an artifact of misapplication.

Its weak point is the anchor. Section 4.1 shows that the measured Jovian homopause coefficient is itself uncertain by nearly an order of magnitude, and that uncertainty propagates through Eq. (4) unchanged.

4. Empirical determinations

4.1 Solar-system homopause occultations

The only direct measurements of K_{zz} at a homopause are solar-system occultations. Festou et al. (1981) is the standard Jovian case: Voyager 2 watched the star Regulus set behind Jupiter’s atmosphere on 9 July 1979, and the absorption in the 910–1200 Å range (H_2 Lyman and Werner bands) and in the 1250–1600 Å range (hydrocarbons) gives density profiles for H_2 , CH_4 and C_2H_6 directly. The heavier species fall off faster than H_2 above the homopause, and where the transition occurs fixes K_{zz} . Their published abstract states that the analysis “yields a value of the eddy diffusion coefficient at the homopause” of

$$K_{\text{homopause}}(\text{Jupiter, equatorial}) = 1.4_{-0.7}^{+0.8} \times 10^6 \text{ cm}^2 \text{ s}^{-1}, \quad (5)$$

consistent, they note, with the value deduced from the $\text{H Ly}\alpha$ and $\text{He } 584 \text{ Å}$ emission data.

The method is the cleanest available: no chemical network, no free efficiency parameter, and the same physics (a species-mass-dependent scale height above a mixing level) that Eq. (2) implements. Its limitations are equally clear: it measures K_{zz} at one level, on one planet, in one region — the paper is explicit that the value is for the equatorial region — and it presupposes that a diffusive description is appropriate, which is the assumption Sect. 2.2 questions.

This number matters here for a reason beyond its own interest. Arfaux & Lavvas anchor Eq. (4) at $K_{zz}^J = 10^7 \text{ cm}^2 \text{ s}^{-1}$, after Koskinen et al. (2010) as recorded in `kzz_literature.md` (that attribution is not verified here). That is a factor $\simeq 7$ above Festou et al.’s measured value. The exoplanet homopause scaling therefore inherits a factor of several before any planet-to-planet extrapolation begins. If the anchor were taken at the Festou value instead, Eq. (4) would give $4 \times 10^6 - 2 \times 10^7 \text{ cm}^2 \text{ s}^{-1}$ for LHS 1140 b, close to the molecular coefficient at the base (Sect. 5.) — i.e. close to no eddy mixing at all in this problem. This is stated as a sensitivity, not as a recommendation: the two Jovian numbers may refer to different levels, regions or epochs, and reconciling them is outside this memo.

4.2 Chemical quenching

The second empirical route is disequilibrium chemistry. A species whose chemical relaxation time t_{chem} is short compared with the mixing time $t_{\text{mix}} \simeq H^2/K_{zz}$ tracks local chemical equilibrium; a species for which the inequality reverses is frozen, or “quenched”, at the abundance it had where $t_{\text{chem}} = t_{\text{mix}}$. The observed abundance therefore measures K_{zz} at the quench level, provided the chemical network is known.

Prinn & Barshay (1977) is the founding application, on CO in Jupiter’s visible atmosphere. Their published abstract states that CO’s presence “is a direct result of very rapid upward mixing from levels in the deep atmosphere where the temperature is about 1100 degrees K and where carbon monoxide is thermodynamically much more stable”, that “as a consequence the observed carbon monoxide mixing ratio is a sensitive function of the vertical eddy mixing coefficient”, and that the inferred coefficient is “about three to four orders of magnitude greater than that in the earth’s troposphere”. The method is now standard on exoplanets: Moses et al. (2011) use it, together with mixing-length estimates from a GCM, to build the K_{zz} profiles for HD 189733 b and HD 209458 b that Koskinen et al. (2013b) and much subsequent work quote.

Its two limitations bear directly on the present problem. The quench level sits at bars to tens of millibars — deep — and the value obtained there has to be extrapolated by many decades to reach $1 \mu\text{bar}$, which is where the plateau assumptions of Moses et al. and of Sect. 3.4 enter. And the tracer has, by construction, a finite chemical lifetime comparable to the mixing time, which is precisely the regime in which Zhang & Showman (2018a,b) find the inferred coefficient to be tracer-dependent (Sect. 2.2).

4.3 Cloud and haze lofting

A third empirical handle gives bounds rather than values. Particles of a given size settle at a computable terminal velocity; if they are observed aloft, K_{zz} must be large enough to keep them there. Ackerman & Marley (2001) turn this into a cloud model by balancing turbulent diffusion against sedimentation with an adjustable sedimentation efficiency f_{rain} ; Koskinen et al. (2013b) use the same reasoning in the other direction on HD 209458 b, arguing that $K_{zz} \gtrsim 10^9 \text{ cm}^2 \text{ s}^{-1}$ is required to prevent cloud settling through the cold trap. The output of this route is a one-sided constraint entangled with an unknown particle size, and it lands at millibar levels, not at $1 \mu\text{bar}$.

4.4 Can He 10830 itself constrain K_{zz} ?

There is a fourth possibility, specific to this problem: the He 10830 line is made from helium that has crossed the homopause, so its strength measures how much helium got through. The scan of LHS1140b/kzz_decision.md, section 3 is exactly that experiment, and it establishes both the sensitivity and its limit.

Table 1: Response of the LHS 1140 b model to He_Kzz, from LHS1140b/kzz_decision.md, section 3. All rows are JFNK steady states at He/H=0.55 with He_diffusion: True, identical in every other respect; the homopause radius is where the run’s own D_{eff} equals its K_{zz} . The last two rows are the diffusion-off reference at the same reservoir composition and the measurement of Cherubim et al. (2026).

K_{zz} [$\text{cm}^2 \text{s}^{-1}$]	homopause [R_p]	$\log_{10} \dot{M}$ [g s^{-1}]	red depth [%]	red EW [% Å]
0	below the base	7.61	9.97×10^{-5}	2.5×10^{-5}
10^6	1.0004	7.68	1.00×10^{-2}	2.27×10^{-3}
10^7	1.0108	7.68	2.57×10^{-2}	5.78×10^{-3}
10^8	1.0282	7.68	2.86×10^{-1}	6.54×10^{-2}
10^9	1.0556	7.73	1.289	3.02×10^{-1}
10^{10}	1.1286	7.84	2.514	6.05×10^{-1}
10^{11}	1.4354	7.84	2.538	6.18×10^{-1}
diffusion off	—	7.76	4.285	1.109
measured (C26)	—	—	$1.24^{+0.22}_{-0.23}$	1.108 ± 0.030

Read as a measurement of K_{zz} , Table 1 says four things.

- *The response is enormous and monotonic below the plateau.* The equivalent width rises by four orders of magnitude between $K_{zz} = 0$ and 10^{10} . In that range the line is a sharp probe.
- *It saturates.* Between 10^{10} and 10^{11} the equivalent width moves by 2%. Above the saturation point the line carries no information about K_{zz} at all, because the composition in the line-forming region has stopped responding (Sect. 5.3).
- *It is degenerate with composition.* The equivalent width measures the metastable column, which is set by the product of the reservoir He/H and the fraction of it that survives the homopause. One observable cannot separate two unknowns. The scan was run at He/H=0.55, a value that was itself fitted in the well-mixed limit, so the scan measures a slice through the degeneracy rather than the degeneracy itself.
- *The wind barely notices.* $\log_{10} \dot{M}$ moves by 0.23 dex over eleven decades of K_{zz} , and not monotonically. The eddy term is a composition control, not a mass-loss control, so the mass-loss rate cannot be used as an independent lever on it.

Breaking the degeneracy needs a second observable with a different dependence on the two unknowns. The natural candidates are a hydrogen line formed in the same region — the ratio of a He and an H diagnostic depends on composition and on separation differently — or a lower-atmosphere composition prior, which is what a Phase-E base.inp

handoff would supply (Sect. 5.5). Neither exists for this planet at present. As things stand the He 10830 line constrains the *product* of reservoir composition and eddy mixing, and quoting a K_{zz} from it alone would be circular.

5. Application: the LHS 1140 b wind base

5.1 The base, and what the eddy term competes against

The run this is for has its base at $p = 1.0 \text{ dyn cm}^{-2} = 1.0 \mu\text{bar}$, $T_0 = 226 \text{ K}$, derived $n_0 = 3.2049 \times 10^{13} \text{ cm}^{-3}$, on a planet with $R_p = 1.730 R_\oplus$, $M_p = 5.60 M_\oplus$ and $g = 1837 \text{ cm s}^{-2}$ (LHS1140b/kzz_literature.md, header). The Banks & Kockarts hard-sphere binary coefficient the operator itself uses,

$$D_{12} = 1.52 \times 10^{18} (1/A_s + 1/A_t)^{1/2} T^{1/2} / n, \quad (6)$$

evaluates at those conditions to $D(\text{H}, \text{He}) = 8.0 \times 10^5 \text{ cm}^2 \text{ s}^{-1}$, and the run's own stage-resolved D_{eff} at the first face agrees, 4.6×10^5 .

The crucial structure is not that number but its gradient. From kzz_decision.md, section 1, D_{eff} rises from 4.6×10^5 at the base to $2.5 \times 10^{10} \text{ cm}^2 \text{ s}^{-1}$ at $1.2 R_p$ and 1.4×10^{11} at $2 R_p$ — nearly five decades over one fifth of a planetary radius, driven by the falling density (a factor 500 in pressure over that interval) and the rising temperature. Against a *constant* K_{zz} this has an immediate consequence:

any K_{zz} above $\sim 10^6$ mixes the base cell, and no K_{zz} below $\sim 10^{11}$ mixes anything at $2 R_p$.

The choice of He_Kzz therefore sets *how far above the base* the homopause sits and nothing else; it never decides whether the base separates.

5.2 The literature at $1 \mu\text{bar}$

Table 2 collects the published values evaluated or extrapolated to the base pressure of this run, with the verification status carried across from kzz_literature.md.

The spread is five decades, and it is not random: it sorts by how hot and how strongly forced the source atmosphere is. The two entries that are not hot-Jupiter extrapolations — the scaling evaluated for this planet, and the one measurement in the table — are also the two lowest. Cherubim et al. (2026), the paper the LHS 1140 b measurement comes from, contains no eddy term, no molecular diffusion and no homopause anywhere in its text or supplement, so it offers no constraint of its own; its H:He is a fitted constant of a homogeneous Parker wind.

5.3 The outer wind, where none of this applies

One result of the scan deserves to be separated from the choice of value, because it is a statement about the physics rather than about the number. Even at $K_{zz} = 10^{11} \text{ cm}^2 \text{ s}^{-1}$ — above every published value in Table 2 — the elemental ratio (He/H)/HeH freezes at 0.43 above $5 R_p$, and at the adopted 10^9 it freezes at 0.16. Since half the $n(2^3S)rdr$ integral lies above $4.2 R_p$, most of the He 10830 line is made in gas that no eddy coefficient reaches. That is why Table 1 saturates.

Table 2: Published K_{zz} evaluated at $1\mu\text{bar}$. “Verified” means the quoted wording and numerals were read from the publisher’s PDF; “ADS full text” means they were read from the arXiv version with key sentences confirmed present in the ADS full-text index of the published article. Full detail, including which sentences, is in LHS1140b/kzz_literature.md.

source	object	$K_{zz}(1\mu\text{bar})$ [cm^2s^{-1}]	inside stated range?	verified?
Parmentier et al. (2013)	HD 209458 b	5×10^{11}	yes	ADS full text
Moses et al. (2011)	hot Jupiters	$10^{11}\text{--}4 \times 10^{11}$	yes, by assumption	ADS full text
Charnay et al. (2015)	GJ 1214 b	$1.8\text{--}7.5 \times 10^9$	no (1.5 dec. above top)	verified
Taylor et al. (2025)	HD 209458 b	10^9	yes (quoted at 10^{-6} bar)	verified
Arfaux & Lavvas (2023)	<i>LHS 1140 b</i>	$2.8 \times 10^7\text{--}1.4 \times 10^8$	yes (plateau)	ADS full text
Blain et al. (2021)	K2-18 b	10^6 nom., $10^5\text{--}10^{10}$	not pressure-resolved	ADS full text
Festou et al. (1981)	<i>Jupiter, measured</i>	1.4×10^6	at the homopause	abstract (Sect. 6.)

The reason is visible in the D_{eff} profile: molecular diffusion overtakes any constant eddy term within a few hundredths of a radius of the base and stays four to five decades ahead of it thereafter. But there is a second reason, and it is conceptual rather than numerical. Above the homopause the atmosphere is not a turbulent fluid at all; it is a transonic outflow in which composition is set by the competition between advection and molecular separation. Carrying a large constant K_{zz} out to $30R_p$ imposes turbulent mixing in a region where the closure of Eq. (1) has no eddy field to describe. The operator’s behavior there is dominated by molecular diffusion, so the imposition is currently harmless; but it means a large He_Kzz should be read as a statement about the base region only, and should never be tuned to fix outer-wind composition.

5.4 The adopted value, and the weakness of its basis

LHS1140b/kzz_decision.md adopts $\text{He_Kzz} = 1.0 \times 10^9 \text{ cm}^2 \text{ s}^{-1}$, on four grounds: it is the only value with a published justification for a model of the same kind at the same pressure (Taylor et al. 2025, verified against the published PDF); it is what EXHALE already uses in examples/14_diffusion and the HD 209458 b Phase-D baselines; it sits between the two literature clusters; and it sits on the steep part of the response rather than the plateau, so its influence is visible to a reader.

Every one of those is an argument from convention or from convenience, and none is an argument from LHS 1140 b. Stated plainly:

- The one estimate derived from this planet’s own gravity and temperature is $2.8 \times 10^7\text{--}1.4 \times 10^8$, one to two decades lower (Sect. 3.4).
- If its Jovian anchor were taken from the measurement rather than from the modelling value, it would be lower again by a factor $\simeq 7$ (Sect. 4.1).
- All three theoretical branches — convective, wave-driven and homopause — predict that a colder, more weakly

forced, higher-gravity atmosphere has a smaller K_{zz} (Sect. 3.). Nothing in the surveyed literature argues the other way.

- The observable moves by a factor $\simeq 9$ in equivalent width between 10^8 and 10^{10} (Table 1).

The honest form for a paper is therefore the adopted value *and* the 10^8 – 10^{10} bracket with the line strengths it gives, which is what `kzz_decision.md` section 5 recommends.

5.5 Three steps toward a determination

In increasing order of cost and of credibility:

1. Report the Arfaux–Lavvas axis alongside the adopted value. No new physics and no new runs beyond the existing scan: quote $\text{He_Kzz} = 10^9$ as the hot-Jupiter convention and 10^8 as the planet-specific homopause scaling, with both line strengths. This costs nothing and removes the main criticism of the present choice, which is not that the value is wrong but that its provenance is invisible.

2. Give K_{zz} a profile instead of a constant — done for the handoff, and measured. The physics of Sect. 3.2 does not produce a constant; it produces growth as

$$K_{zz}(P) = K_{\text{ref}} \left(\frac{P}{P_{\text{ref}}} \right)^{-\alpha}, \quad \alpha = 1/2, \quad (7)$$

up to a saturation level, and a turnover above it.

A radially varying K_{zz} has not been hypothetical in the code since Phase E: the operator reads a cell array `kzz_cell` (`src/modules/init/parameters.f90`), which a `Lower atmosphere profile: file` fills by interpolating its own $K_{zz}(p)$ column onto the grid (`src/modules/files_IO/lower_atmosphere_profile.f90`) and which is filled uniformly from He_Kzz otherwise. What was missing was a way to *state* Eq. (7); that is now there, on the handoff side. `eddy_diffusion_coefficient` (`src/utls/lower_profile_schema.py`) takes `--kzz-power` α with its amplitude `--kzz-ref` K_{ref} and the pressure that amplitude is stated at, `--kzz-ref-bar` P_{ref} , and writes the resulting column into the profile. In `photochem_to_lower_profile.py` the stated coefficient is the one the chemistry is *solved* on and not only the one the file reports, so the composition handed over is the composition that $K_{zz}(p)$ produces; in the VULCAN cross-check arm the chemistry is already finished, so a stated coefficient changes only what is handed over. With neither option given each adapter keeps whatever $K_{zz}(p)$ its own solution carried, and the `solution_id` of a constant- K_{zz} run is unchanged.

What the pressure form was then measured to be worth. On LHS 1140 b, at the two reservoirs of the flux-closed ladder that straddle the measurement, Eq. (7) with $\alpha = 1/2$ and $K_{\text{ref}} = 10^9$ anchored at the top of the profile (1.1×10^{-8} bar) — which leaves the value the wind inherits above the file unchanged and gives 1.05×10^8 at the $1 \mu\text{bar}$ match, 1.05×10^5 at 1 bar and 2.56×10^4 at the 16.73 bar deep boundary — moves the He 10830 equivalent width by +0.28% when only the chemistry sees it and +0.10% when the wind sees it too (He/H = 11.11 arm), and by +0.61% and –0.15% on the He/H = 2.09 arm. The FWHM and $\log_{10} \dot{M}$ do not move at all. The reason is visible one step

earlier: the elemental ratios at the match, which with the temperature are what EXHALE consumes from the profile, agree with the constant- K_{zz} ones to 1.2% or better (X_{He} 0.9985, X_{C} 0.990, X_{N} 0.988, X_{O} 0.982–0.989), even though the *carriers* move by up to four decades and nitrogen changes carrier outright from NH_3 to N_2 . Re-solving the same profile from a different seed moves the equivalent width by 1.5%, an order of magnitude more than the eddy form does. Measurement and runs: LHS1140b/exhale/kzz_power/, recorded in LHS1140b/kzz_decision.md section 9 and docs/lhs1140b_exhale_vs_pwinds.tex, “What the constant K_{zz} carries”.

The real remaining gap is the two-branch law, not the profile. Three things are still absent, in decreasing order of how much they matter.

(i) *There is no convective branch.* Equation (7) is the *stratified* law, and on LHS 1140 b the climate solve puts the tropopause at 2.636 bar; below it the mixing-length form of Eq. (3) applies and K_{zz} should flatten or grow with depth rather than collapse. Running the single power law to 16.73 bar gives $1.05 \times 10^5 \text{ cm}^2 \text{ s}^{-1}$ at 1 bar, at or below the floor of the modelling literature — one and a half to two and a half decades under the 3×10^6 – 3×10^7 of Charnay et al. (2015), 3.7 decades under the 5×10^8 of Parmentier et al. (2013), and equal to the 10^5 minimum Ackerman & Marley (2001) impose by hand. What is needed is the pair: a wave law above the radiative–convective boundary joined to a mixing-length law below it, with the boundary taken from the climate solve that is already in the same adapter.

(ii) *There is no ceiling.* An unbounded $P^{-1/2}$ carried far above the level its amplitude was set at reaches absurd values and contradicts Lindzen’s own turnover result. The option is deliberately confined to the handoff, where the file’s own top bounds it and where EXHALE gives every cell above that top the top level’s value. That bound is also what limits the measurement above on the wind side: the LHS 1140 b column reaches only $1.0091 R_p$ past the match, so the coefficient the wind saw was changed in a shell shallower than the $1.056 R_p$ homopause and not where helium settles. A caller who anchors the same law at the match instead of at the top raises the coefficient over the whole wind domain by the same factor and should read that as a change of magnitude, not of form.

(iii) *A run with no profile file still has only the constant He_Kzz.* Nothing about Eq. (7) has been put into binary_element_diffusion.f90 itself.

None of this repairs Sect. 5.3: D_{eff} rises faster than $P^{-1/2}$ in this wind (roughly as $T^{1/2}/n$, and T rises steeply through the same interval), so no pressure form of the eddy coefficient reaches the region where most of the line is made. The gain is not a stronger line; it is that the base region is described by the physics the literature values were derived from. And the measurement above says what the gain is worth on the observable, which is very little — while noting that the anchor’s own value at the match, 1.05×10^8 , is better founded for this planet than the adopted 10^9 , since it falls inside the 2.8×10^7 – 1.4×10^8 of Eq. (4).

3. A determination, by GCM or by simultaneous fit. Either measure K_{zz} for this planet the way Sect. 3.3 describes — a GCM of a temperate, helium-rich, tidally locked super-Earth with passive tracers, which does not currently exist — or fit K_{zz} jointly with the reservoir composition against the measured line, accepting that a single observable cannot separate the two (Sect. 4.4) and that the fit would therefore need a second constraint. The Phase-E route is a partial version of the second, and it is now built: a Lower atmosphere profile: file carries the composition *and* $K_{zz}(p)$ from the photochemical model, so both reach the wind from one calculation instead of two hand-set numbers. The scalar

`Kzz_base` key of `read_base_inp` remains for a run handed over by scalars alone, and is refused when a profile is in use (`refuse_scalar_key`, `src/modules/files_IO/input_read.f90`).

None of the three makes K_{zz} a derived quantity of EXHALE. It stands for turbulence and wave breaking that a one-dimensional wind model does not resolve, so it is stated, not computed — which is why the code’s default is 0 rather than an inherited number.

6. Verification status of the new citations

The citations below were not in `LHS1140b/kzz_literature.md` and were checked for this memo. Publisher PDFs could not be retrieved for three of the four: the ADS link gateway returned HTTP 404 for PUB_PDF on the two *JGR* papers and on the *Science* paper, and HTTP 400 on the *ApJ* paper. Where the PDF was unavailable, quoted wording was checked by (i) the abstract of the published article as indexed by ADS and (ii) full: ". . ." phrase queries against the ADS full-text index of the published article, which is the same procedure used in `kzz_literature.md`. All four papers are full-text indexed.

Table 3: New citations and how each was checked.

reference	bibcode	edition read	status
Lindzen (1981)	1981JGR...86.9707L	published abstract	partial
Ackerman & Marley (2001)	2001ApJ...556..872A	arXiv:astro-ph/0103423	partial
Prinn & Barshay (1977)	1977Sci...198.1031P	published abstract	partial
Festou et al. (1981)	1981JGR...86.5715F	published abstract	partial

Lindzen (1981). Every sentence quoted in Sect. 3.2 is from the abstract of the published article as indexed by ADS. Two of them were additionally confirmed present in the ADS full-text index of the published article: full:"turbulence in the upper mesosphere arises from the unstable breakdown of tides and gravity waves" and full:"prevent the growth of wave amplitude with height" each return the paper; so does full:"eddy coefficients may peak". **Caveat:** the full-text index of this paper appears to be partial or OCR-degraded — full:"eddy diffusion", full:"Richardson number" and full:"vertical wavelength" all return nothing, while full:"eddy coefficient", full:"critical level" and full:"momentum deposition" return the paper — so absence of a phrase is not evidence here. Lindzen’s saturation formula for the eddy coefficient is deliberately *not* written out in this memo: it was **not verified against the published text**, and the physical argument in Sect. 3.2 is stated in words that the abstract supports.

Ackerman & Marley (2001). Read from the arXiv PDF (astro-ph/0103423, retrieved through the ADS EPRINT_PDF gateway). Eq. (3) is their eq. (5), with the attribution to Gierasch & Conrath (1985) in their own text. The following were confirmed present in the ADS full-text index of the published *ApJ* article: “is assumed to be the same as that for heat as derived for free convection”; “the constant coefficient scaling the eddy diffusion coefficient is

only loosely constrained by observations”; “the mixing length is typically assumed to be the pressure scale height”; “the convective velocity scale from mixing-length theory”; “we also assume an eddy diffusion coefficient no less than a prescribed minimum value”. **Caveat:** the composite phrase “which represents residual turbulence due to breaking buoyancy waves” returns nothing, while “residual turbulence” and “breaking buoyancy waves” each return the paper. The most likely cause is that an inline equation splits the sentence in the published typesetting, but the composite wording is **not verified against the published text** and is quoted here only as its two verified fragments. The numerals ($K_{\min} = 10^5$, $H = L = 20$ km, $K = 2 \times 10^8$ cm² s⁻¹, $w_* = 1.1$ m s⁻¹, $T_{\text{eff}} = 124$ K, prefactor 1/3) are read from the arXiv version and are **not verified against the published text**.

Prinn & Barshay (1977). All three quotations in Sect. 4.2 are from the abstract of the published *Science* article as indexed by ADS. Two were additionally confirmed present in the ADS full-text index of the published article: full:“sensitive function of the vertical eddy mixing coefficient” and full:“three to four orders of magnitude greater than that in the earth”. No numerical value of K_{zz} from this paper is quoted in this memo, because none was verified: full:“eddy diffusion coefficient” and full:“quench” both return nothing for this bibcode, so the paper uses different wording and the body could not be read.

Festou et al. (1981). The measured homopause coefficient $1.4^{+0.8}_{-0.7} \times 10^6$ cm² s⁻¹, the equatorial qualifier, the consistency with the H Ly α and He 584 Å determination, the occultation date, the star, and the wavelength ranges are all from the abstract of the published article as indexed by ADS. The phrase full:“yields a value of the eddy diffusion coefficient at the homopause” returns the paper in the ADS full-text index of the published article, so the abstract wording is present in the article itself. The body of the paper was not read.

Not verified. Two further attributions are carried over from `kzz_literature.md` without independent checking here and are marked in the text: the anchor $K_{zz}^J = 10^7$ cm² s⁻¹ that Arfaux & Lavvas (2023) attribute to Koskinen et al. (2010) (Sect. 4.1) — the specific Koskinen et al. (2010) paper was not identified — and the identification of Eq. (3) with eq. (16) of Charnay et al. (2015).

7. References

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Companion documents in this tree. LHS1140b/kzz_literature.md (the survey, with the verification status of every published number); LHS1140b/kzz_decision.md (base conditions, the D_{eff} structure, the He_Kzz scan, the adopted value and the alternatives); docs/binary_diffusion_design.md (the operator, sections 2.3 and 9.4); docs/lhs1140b_exhale_vs_pwinds.tex (the He 10830 comparison the composition was fitted against).